Prediction of Turbulent Flow Over Rough Surfaces Using a Force Field in Large Eddy Simulation

Jie Cui
Virendra C. Patel
Ching-Long Lin

IIHR-Hydroscience and Engineering
and Department of Mechanical Engineering,
The University of Iowa,
Iowa City, IA 52242-1585

A force field model to simulate turbulent flow over a surface with arbitrary roughness is described. A given roughness is decomposed into resolved and subgrid-scale roughness, conceptually similar to the flow decomposition in large eddy simulation (LES). For a given flow and Reynolds number, a Cartesian grid is selected to satisfy LES requirements. This grid determines the geometric features of the roughness that are formally resolved. The force field used to represent this resolved roughness is determined during the LES solution process, without any empirical input. The subgrid roughness that is not resolved is modeled by a random force distribution in which a drag coefficient is specified. Use of this approach to model surface roughness is demonstrated by calculations of the flow in a duct with a wavy wall with superimposed fine-grain roughness.
[DOI: 10.1115/1.1524587]

1 Introduction

Turbulent flow over rough surfaces is hardly a new research topic. Nikuradse [1] experimented on sandgrain roughness in 1933. Subsequently, numerous experimental studies have been conducted to understand flow phenomena associated with different types of surface roughness and parameterize their effect on resistance, velocity distribution, and transport of heat and mass. Computational fluid dynamics (CFD) has not significantly advanced the understanding of this problem because it has relied mostly on concepts of equivalent sandgrain roughness and empirically determined roughness functions. In Reynolds-averaged Navier-Stokes (RANS) simulations, it is common to avoid resolution of the flow in the roughness layer and wall region adjacent to the roughness by employing the method of wall functions. An alternative, again in the framework of RANS simulations, is the so-called discrete element approach, in which roughness due to a regular array of discrete two or three-dimensional elements, such as ribs, cones, and hemispheres, is considered by adding a form drag term in the momentum equation, and under certain circumstances, accounting for the blockage effect of the roughness elements in the continuity equation, [2–7]. Patel [8] has made a review of such approaches.

Recently, roughness modeling has been extended to large-eddy simulation (LES) and direct numerical simulation (DNS). For example, Lin et al. [9] used LES to investigate surface roughness effects on flow structures in a neutrally stratified planetary boundary layer. The surface roughness treatment was similar to the wall-function approach in RANS models. Although not specifically concerned with roughness, Patton et al. [10] reported LES of turbulent flow around multiple porous windbreaks within a wheat canopy under neutral stability conditions. An empirical term representing drag of the windbreak was added to the momentum equation as in previous RANS simulations.

In the DNS of turbulent channel flow by Miyake et al. [11], a rough wall was modeled by a distribution of zero-volume ''virtual'' cones. A drag force d_i per unit length of the form

$$d_i = C_d \frac{1}{2} \rho u^2 \frac{u_i}{|u|} \frac{A}{V} \tag{1}$$

was added at the grid points inside these ''virtual'' cones, C_d being the drag coefficient, A the frontal area, and V the volume of the cone. The cone height was generated randomly with a mean value between 15 and 30 wall units, and a value of 0.5 was prescribed for C_d. The well-known effect of surface roughness in the logarithmic velocity distribution was reproduced, and the authors claimed that the effect of roughness on the eddy structure was also simulated. However, the results depend on the chosen value of C_d.

2 Present Work

The present work is inspired by some recent LES and DNS computations using ''body-force'' or ''immersed-boundary'' methods to represent very complex geometries. An example of this is the recent LES of Verzicco et al. [12] who simulated the flow in a piston-cylinder-valve assembly. The simulations were carried out in a Cartesian grid with the geometry represented by a body-force field to enforce the known (zero or prescribed) velocity within solid regions not occupied by the fluid. Although these and similar solutions were not concerned with surface roughness per se, their approaches can be extended to develop a general and powerful model of turbulent flow over rough walls. This is the purpose of the present paper.

The boundary body-force model of Verzicco et al. [12] is attractive for several reasons. First, an orthogonal Cartesian grid is used to represent complex geometry, avoiding a body-fitted (and/or deforming) grid. Second, the computation scheme remains simple and efficient, and does not introduce additional stability limits. Third, there are no parameters in the model that need to be determined beforehand by reference to experimental data because the body force magnitude is obtained internally during the numerical solution. This makes the model much more general than those involving empirical inputs. Finally, in this model, steady and moving boundaries can be treated in the same way, which is convenient for time-dependent problems.

The numerical method and decomposition of roughness into resolved and subgrid components are described in Section 3. Turbulent flow in a channel with a wavy wall is used as an example. This flow is first simulated using the traditional approach of a body-fitted grid and the results are used in Section 4 to validate the numerical method and to establish a reference to evaluate the roughness decomposition and representation by body-force distribution presented in Section 5. Section 6 describes the treatment of resolved-scale roughness, while Section 7 describes modeling of subgrid roughness. Section 8 summarizes the major results and conclusions.

Contributed by the Fluids Engineering Division for publication in the JOURNAL OF FLUIDS ENGINEERING. Manuscript received by the Fluids Engineering Division Dec. 28, 2001; revised manuscript received Aug. 6, 2002. Associate Editor: T. B. Gatski.

 Copyright © 2003 by ASME

Downloaded From: http://fluidsengineering.asmedigitalcollection.asme.org/pdfaccess.ashx?url=/data/journals/jfega4/27181/ on 06/07/2017 Terms of Use: http://www.asme.org/al

3 Numerical Method and Roughness Decomposition

The three-dimensional, unsteady, incompressible, filtered continuity and Navier-Stokes equations, (2) and (3), respectively, are solved with a dynamic subgrid-scale model (DSM):

$$\frac{\partial \overline{u_i}}{\partial x_i}=0 \tag{2}$$

$$\frac{\partial \overline{u_i}}{\partial t}+\frac{\partial}{\partial x_j}(\overline{u_i u_j})=-\frac{1}{\rho}\frac{\partial \overline{p}}{\partial x_i}+\nu\left(\frac{\partial^2 \overline{u_i}}{\partial x_j \partial x_j}\right)-\frac{\partial \tau_{ij}}{\partial x_j}+f_i. \tag{3}$$

Here $i=1$, 2, and 3, $\overline{u_i}$ are the resolved velocity components, $\overline{p}$ is pressure, ν is kinematic viscosity, f_i is the body force, and

$$\tau_{ij}=\overline{u_i u_j}-\overline{u_i}\,\overline{u_j} \tag{4}$$

is the subgrid-scale stress representing the effect of small-scale motions.

In DSM, the model coefficients are computed dynamically as the calculation progresses rather than being prescribed a priori. Two filters are introduced: the grid filter and the test filter. A test filter twice as large as the grid filter is usually chosen and is used in this paper. By applying the test filter to the equations, the subgrid-scale stresses on the test level, or subtest-scale stresses can be constructed. Then subgrid-scale stresses are parameterized by an eddy viscosity model:

$$\tau_{ij}-\frac{\delta_{ij}}{3}\tau_{kk}=-2C\overline{\Delta}^2|\overline{S}|\overline{S}_{ij} \tag{5}$$

where

$$|\overline{S}|=(2\overline{S}_{ij}\overline{S}_{ij})^{1/2},\quad \overline{S}_{ij}=\frac{1}{2}\left(\frac{\partial \overline{u_i}}{\partial x_j}+\frac{\partial \overline{u_j}}{\partial x_i}\right),$$

δ_{ij} is the Kronecker delta, $\overline{\Delta}=(\Delta x_1 \Delta x_2 \Delta x_3)^{1/3}$, Δx_i are the mesh distances, and C is a model constant. To obtain a single coefficient from these independent equations, C is assumed to remain unchanged between these two filters, [13–15]. Then the model constant C can be determined from

$$C=\frac{1}{2}\frac{L_{ij}M_{ij}}{M_{ij}M_{ij}} \tag{6}$$

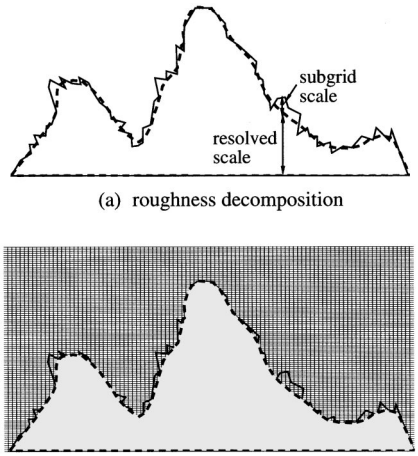
where

$$L_{ij}=\overline{\overline{u_i u_j}}-\overline{\overline{u_i}}\,\overline{\overline{u_j}}\quad\text{and}\quad M_{ij}=\overline{\Delta}^2|\overline{\overline{S}}|\overline{\overline{S}}_{ij}-\bar{\bar{\Delta}}^2|\overline{\overline{S}}|\overline{\overline{S}}_{ij}$$

(double overbar represents test-scale resolved quantities) can be directly computed from the resolved motion.

The finite volume LES numerical code used here was developed by Zang et al. [16]. In this method, the governing equations in general curvilinear coordinates are discretized on a nonstaggered grid. A fractional-step method is employed and the pressure Poisson equation is solved with multigrid acceleration. Time marching is semi-implicit with formal accuracy of second order in both space and time. Zang et al. [16] verified and validated the code in various flows, including decaying vortices, lid-driven flow in two-dimensional, three-dimensional and polar cavities, flow over a backward-facing step, and flow in a curved square duct.

Analogous to the basic idea of LES and the dynamic subgrid-scale model, a general approach to treat surface roughness, able to handle arbitrary roughness, is proposed. An arbitrary roughness, shown schematically in Fig. 1(a), is decomposed into two parts, a "resolved-scale" roughness and a "subgrid-scale" roughness. Since the large-scale roughness determines the most energetic flow features that vary significantly from problem to problem, it is difficult to construct a model that will work universally for all possible roughness geometry. Here we adopt the approach of Verzicco et al. [12] for the resolved-scale roughness. The length scale of the subgrid-scale roughness, as the name suggests, is much smaller than that of the resolved roughness. The subgrid-scale roughness makes a small contribution to momentum, mass and energy transport, and is expected to have common features in various problems. This portion of the roughness is modeled following the work of Miyake et al. [11].



(a) roughness decomposition

(b) implementation of new force-field model

Fig. 1 Decomposition of arbitrary roughness and application of body-force model

A Cartesian grid is used for the given roughness geometry (Fig. 1(*b*)). Although the figure shows a two-dimensional geometry, the basic method is applicable to any geometry. The required grid will depend on the roughness configuration as well as the Reynolds number, as is the case in all LES applications. The grid should be fine enough to resolve the basic large-scale features of the flow as well as the roughness. The subgrid-scale roughness not resolved by the chosen LES grid is modeled as a random roughness, similar to the "virtual" cones in Miyake et al. [11]. The average length scale needs to be prescribed. This is a purely geometric parameter and can be obtained from the roughness geometry after the resolved-scale roughness is subtracted.

4 Flow in a Wavy-Wall Channel: Test Problem

Consider turbulent flow in a channel with plane wall on the top and a sinusoidal wavy wall on the bottom (Fig. 2). The wave profile is given by

$$h=a\sin\left(\frac{2\pi}{\lambda}x-\frac{\pi}{2}\right) \tag{7}$$

where h is the vertical height with a zero mean value, a is the wave amplitude, x is the streamwise coordinate, and λ is the wave length (Fig. 3). There is no variation in the spanwise direction,

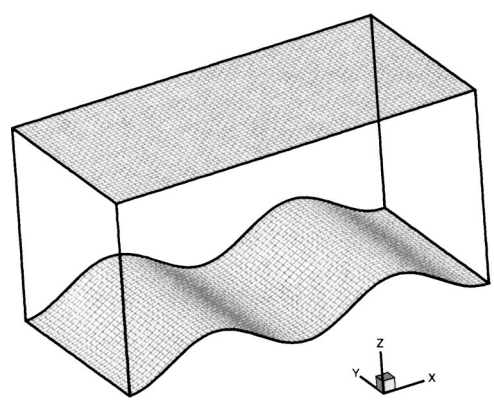


Fig. 2 Geometry of channel flow with wavy wall

Downloaded From: http://fluidsengineering.asmedigitalcollection.asme.org/pdfaccess.ashx?url=/data/journals/jfega4/27181/ on 06/07/2017 Terms of Use: http://www.asme.org/al

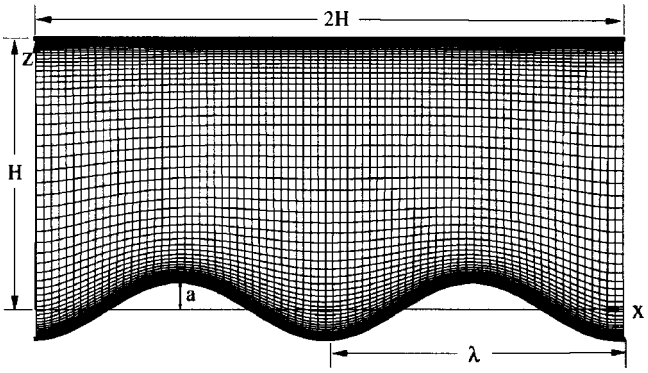


Fig. 3 Body-fitted computational grid in the wavy channel

i.e., the wavy wall is two-dimensional. In the test problem, the wavelength λ is equal to the mean channel height H, the wave steepness, $2a/\lambda$, is 0.20, and the Reynolds number (Re), based on the bulk velocity (U_b) and half-channel height ($H/2$), is 10,000.

As noted earlier, this flow problem is first solved as in conventional LES, using a body-fitted grid. The computational domain contains two waves in the streamwise direction. The spanwise and vertical dimensions are H. A grid with $82\times42\times66$ points in the streamwise (x), spanwise (y), and normal (z) directions is shown in Fig. 3. The grid points are more closely spaced near the two walls, but the grid spacing in the longitudinal and spanwise direction is uniform.

A mean streamwise pressure gradient is imposed to drive the flow, and is carefully adjusted to obtain Re=10,000. Periodic boundary conditions are imposed in the streamwise and spanwise directions, and no-slip boundary conditions are applied at the top and bottom walls. The friction Reynolds number for the upper wall Re_τ is around 570. In terms of wall units, the mesh spacing is $\Delta x^+\approx27$, $\Delta y^+\approx27$, and at least 1 for Δz^+.

As part of a validation study of the numerical method, time and spanwise-averaged mean streamwise velocity profiles at six representative locations within one wave are compared with the experimental data of Buckles et al. [17] in Fig. 4. At the wave crest, $x/\lambda=0.0$ (Fig. 4(a)), the flow is attached, and experiences strong favorable pressure gradient (acceleration) following reattachment on the previous wave. The flow accelerates rapidly in a short distance. The numerical prediction is in good agreement with the measurements except in a region close to the measured velocity maximum which occurs farther from the top wall than predicted by the LES simulation. This difference between the experiment and simulation is seen in the profiles at all streamwise locations. It may be due to the three-dimensional effects in the experiment in which the channel aspect ratio (width-to-height) was about 12:1. In the LES calculations by Henn and Sykes [18], the same trend was also observed. Henn and Sykes argued that the discrepancies could be due to the measurements, since up to 3% discrepancy in the integrated mass flux was reported from the data of Buckles et al. [17]. At $x/\lambda=0.1$, the LES profile shows a small negative velocity indicating separation, though the experiment gives only positive velocities. Thus, an earlier separation is predicted by the numerical method. Farther downstream, at $x/\lambda=0.3$, the pressure gradient is adverse and the flow separates. The reversed flow is well predicted, demonstrating success of LES in simulation of separated flow. At $x/\lambda=0.5$, the wave trough, flow is still separated. The reversed flow velocity is over estimated. Further downstream, the flow encounters a favorable pressure gradient, the near-wall flow accelerates and wall shear stress begins to increase. At $x/\lambda=0.7$, the uphill, the flow reattaches. The LES profile shows a slightly negative velocity at this location while the mea-

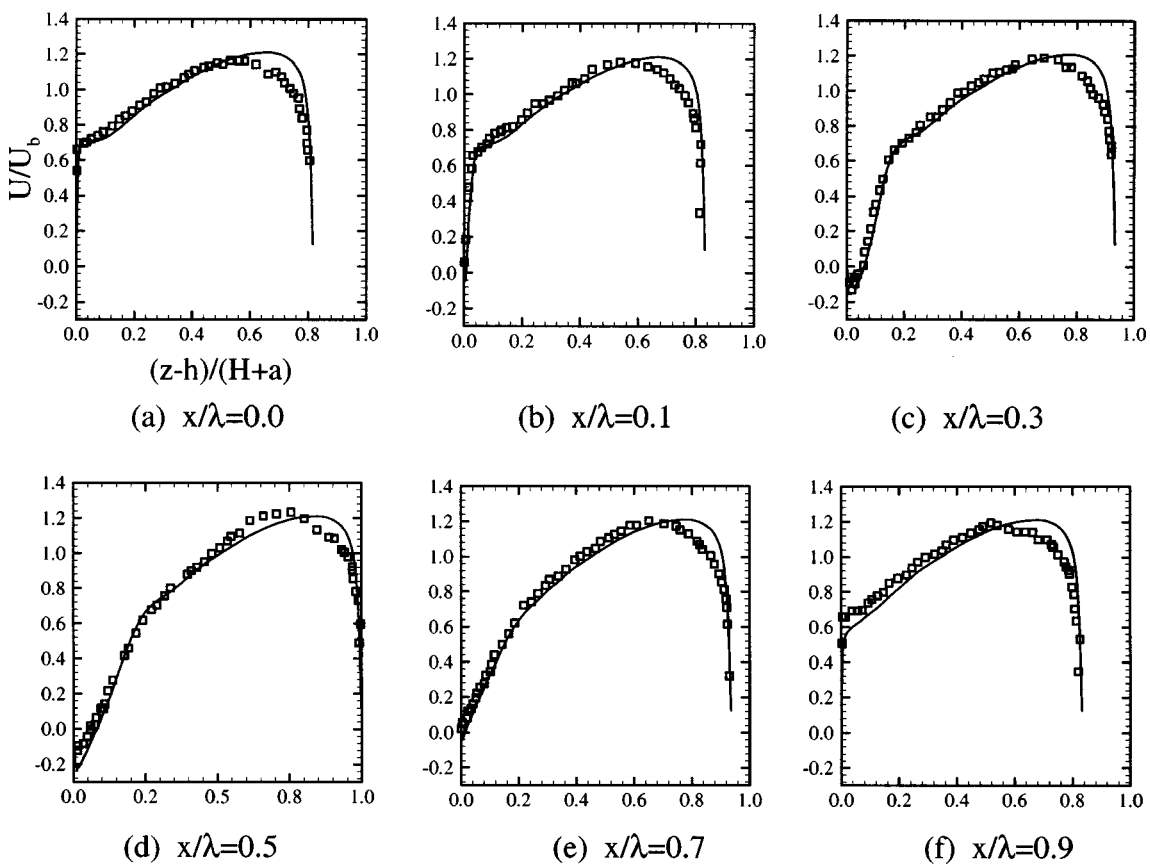


Fig. 4 Comparisons of mean streamwise velocity profiles with experimental data at selected locations

Downloaded From: http://fluidsengineering.asmedigitalcollection.asme.org/pdfaccess.ashx?url=/data/journals/jfega4/27181/ on 06/07/2017 Terms of Use: http://www.asme.org/al

Table 1 Resistance in channel flow with wavy roughness

LES Solution	Resistance Components			Imposed Force
	Lower Boundary		Upper Boundary	
Direct solution with surface-fitted grid	Pressure	Friction		
	0.0415	0.00076	0.0067	
	Total:		0.0489	0.049
	Lower Boundary		Upper Boundary	
Resolved-scale roughness and body-force	Pressure	Friction		
	0.0416	0.00064	0.007	
	Total:		0.0492	0.049
	Lower Boundary		Upper Boundary	
Sub-grid-scale roughness and random body-force model	Pressure	Friction		
	0.0403	-0.00022	0.0071	
	Total:		0.0472	0.049

Note: Forces are per unit width of channel and are normalized by $\rho U_b^2 H$

surements do not, indicating a more downstream reattachment point. Finally, results in the last section, $x/\lambda = 0.9$, in a region of strong favorable pressure gradient, are very similar to those at the wave crest.

To check whether the size of the simulation domain and spanwise grid spacing contribute to the discrepancies with the experimental data, another run was performed on a larger domain and finer grid. The domain was extended to $3H$ in the streamwise direction and the number of spanwise grid points was doubled. The resulting domain was $3H \times H \times H$ and the grid was $122 \times 82 \times 66$. The difference between the results from the two meshes is quite small (both results show nearly identical mean velocity distribution, and that about 7% of the turbulence kinetic energy is carried by the subgrid-scale model). Therefore, it is believed that the majority of the error is due to numerical issues.

Considering the wave train as a surface roughness, albeit of regular shape, the simulation results may be used to calculate the pressure and friction components of resistance to the flow in the channel by integration along one wave. The first set of entries labeled direct solution in Table 1 summarizes the calculated resistance components. The last column in the table shows the pressure force that was imposed. It is found that the friction resistance from the lower wall is negligible due to the large separation zone in the valley behind the crest. The wavy wall generates a large pressure resistance. The contribution from the top wall only contains frictional resistance and is much smaller in comparison with the pressure resistance from the lower wall. The pressure force imposed to drive the mean flow balances the total resistance from lower and upper surfaces, further validating the numerical method and indicating sufficient sampling times to obtain the time-averaged quantities from the LES data. Detailed comparisons with experimental data with respect to other turbulent flow quantities and validation of the numerical method can be found in Cui et al. [19].

5 Modeling Resolved-Scale Roughness With Body Force

The time discretization of Eq. (3) is written as

$$\overline{u_i}^{n+1} - \overline{u_i}^{n} = \Delta t \left(-\frac{\partial \overline{u_i u_j}}{\partial x_j} - \frac{1}{\rho}\frac{\partial \overline{p}}{\partial x_i} + \nu \frac{\partial^2 \overline{u_i}}{\partial x_j \partial x_j} - \frac{\partial \tau_{ij}}{\partial x_j} + f_i \right). \tag{8}$$

To impose $\overline{u_i}^{n+1} = 0$ on points inside the roughness, the body force f_i must be

$$f_i = -\frac{\overline{u_i}^{n}}{\Delta t} + \frac{\partial \overline{u_i u_j}}{\partial x_j} + \frac{1}{\rho}\frac{\partial \overline{p}}{\partial x_i} - \nu \frac{\partial^2 \overline{u_i}}{\partial x_j \partial x_j} + \frac{\partial \tau_{ij}}{\partial x_j}. \tag{9}$$

This body force is determined internally as the solution progresses, and requires no empirical input. The body-force model may be applied to arbitrary roughness, provided a fine enough Cartesian grid resolves the roughness elements. The channel flow with wavy roughness described in the previous section is recalculated using this model, and two sets of results are compared to assess the viability of this simplified approach to modeling roughness.

Figure 5 shows the Cartesian grid for the same computational domain as before with the same number of grid points in the three directions. In the region occupied by the waves, namely, $-0.1 < z/H < 0.1$, this grid is not as fine as that used for the direct calculations of the previous section. In fact, the chosen grid defines the level of resolution of the actual surface and the scale of the subgrid roughness to be modeled.

The mean velocity vectors obtained with the new LES are shown in Fig. 6. The velocities within the waves are now blocked out by the body force. The velocity overshoot at the wave crest and the separation zone in the valley are reproduced. The flow

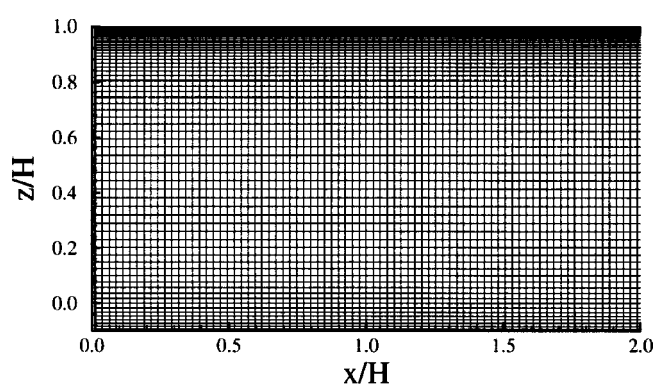


Fig. 5 Computational grid for channel flow over wavy roughness with body-force model

Downloaded From: http://fluidsengineering.asmedigitalcollection.asme.org/pdfaccess.ashx?url=/data/journals/jfega4/27181/ on 06/07/2017 Terms of Use: http://www.asme.org/al

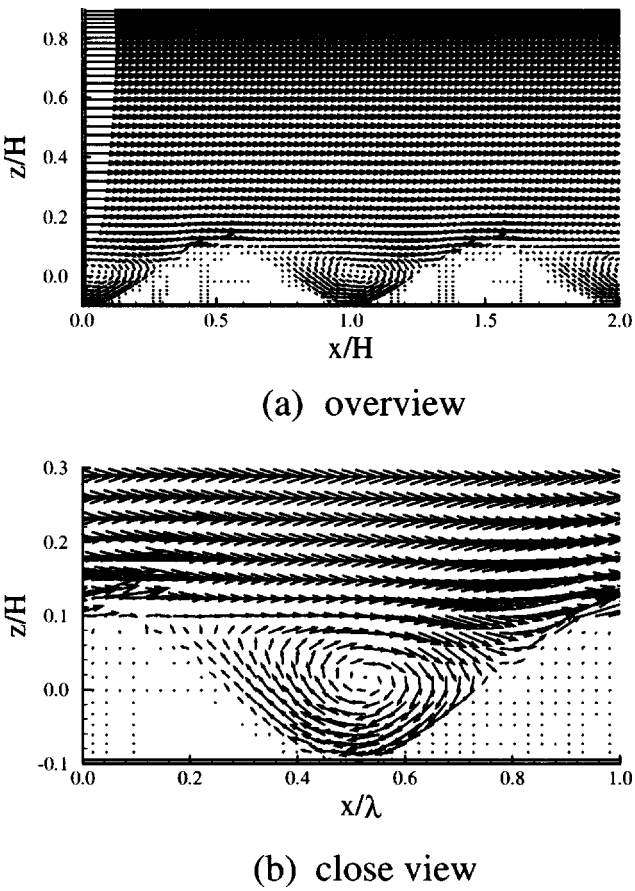


Fig. 6 Velocity vectors for channel flow over wavy roughness using body-force model

field is interpolated onto the body-fitted grid used in Section 4, so that direct comparisons between the two results can be made. In Fig. 7, similar to Fig. 4, time and spanwise-averaged mean streamwise velocity profiles using the two approaches are compared with the experimental data of Buckles et al. [17] at six representative locations. Figures 8 and 9 compare the profiles of wave-average mean velocity ($\overline{U}/U_b$) and turbulent kinetic energy ($\overline{k}/U_b^2$). Wave averaging is done by summing up the physical parameters, already time and spanwise averaged, along the streamwise grid lines, and dividing by the number of grid points on that line. It is seen that the two sets of results match well except close to the wave surface. The difference in the two simulations is due to the approximation made in the treatment of the physical shape of the boundary in the body-force model. No interpolation procedure is taken to scale the forcing at points near the curved surface and the body force is smoothed implicitly by the spatial discretization. The agreement is expected to improve with grid refinement, leaving smaller subgrid roughness effects. As noted by Fadlun et al. [20], improved results could be obtained also by using interpolation to scale the body force at points close to the curved surface.

A particularly interesting result is shown in Fig. 10 where the wave-averaged mean velocity profile is plotted in the usual semi-logarithmic coordinates. Here, $\overline{U}^+ = \overline{U}/\overline{U}_\tau$ and $\overline{z}^+ = \overline{U}_\tau \overline{z}/\nu$, where $\overline{U}_\tau$ is the "friction velocity" based on the equivalent average shear stress on the wave, including the pressure and friction components, and $\overline{z}$ is distance measured from the "virtual origin" defined as the position where the wave-averaged velocity is zero. This plot enables the waves on the bottom wall to be viewed as roughness, and the flow features to be evaluated in the framework of surface roughness effects. It is observed that the two LES simulations, with the wavy surface resolved by a body-fitted grid and represented by a body-force field, give essentially the same result for $\overline{z}^+ > 500$. In other words, the body-force model reproduces the basic effect of roughness observed in the downward shift in the logarithmic law from the smooth-wall flow. The disagreement between the two solutions in the near-wall region, occupied by the roughness itself and the layer just beyond the roughness, is to be expected in view of the approximations made in the treatment of the roughness geometry (the chosen grid) and implementation of the body-force model.

The pressure and friction components of resistance calculated from interpolated surface pressure and shear stress are compared with those of the previous solution in Table 1. The close agreement of the forces calculated by the two methods is a measure of the success of the body-force model.

6 Modeling Subgrid-Scale Roughness by Random Force Field

As depicted in Fig. 1, use of a Cartesian grid to describe the wavy surface leaves the subgrid-scale roughness that has to be modeled. A practical example of this type of decomposition is the flow over a mountainous terrain, where the grid may be sufficiently dense to resolve the mountains and the valleys, but not the forests and buildings. This subgrid-scale roughness is most likely random and can be taken into account using the approach proposed by Miyake et al. [11]. This is demonstrated by taking the case of the wavy wall, assuming that the surface shape left unresolved by the chosen grid is random and with a mean height of 10% of the wave amplitude (i.e., $0.1a$). Following Miyake et al. [11], every grid point within the roughness element height contributes a body force, as in Eq. (1), with $C_d = 0.2$. Miyake et al. [11] used $C_d = 0.5$ for plane channel flow. In the present application, this value required very small time-steps for stability, making the computation very time-consuming. A similar problem was reported by Goldstein et al. [21] when a similar body-force model was applied to complex geometry. It should be pointed that the boundary body-force model [12] adopted for the resolved roughness in the current study does not impose additional stability requirements, while the random force model used for the subgrid-scale roughness discussed in this section does.

The effect of adding the random subgrid-scale roughness is shown in the mean velocity profile in Fig. 11. With the same imposed mean pressure gradient, addition of the "subgrid" roughness decreases the mean velocity. When plotted in logarithmic format, the subgrid-scale roughness increases the downward shift in the velocity profile. It is significant that much of the roughness effect is already captured by the resolved-scale LES, and the subgrid-scale roughness makes a small contribution. For many applications, therefore, it may be sufficient to obtain only a resolved-scale solution. Alternatively, a sufficiently fine grid would obviate the need to add a sub-grid component.

The resistance components with the sub-grid roughness added to the waves are summarized in Table 1. The pressure resistance decreases slightly presumably due to a smaller separation region and the friction component remains negligible. The total resistance from the bottom and top walls is less than the imposed mean pressure drop. The difference is due to the resistance added by the random subgrid roughness elements. In this case, the resistance generated by the subgrid-scale roughness is about 4% of the total resistance.

In the application of the random body-force model, the mean height of the roughness elements is the same as that obtained from the "real" subgrid-scale roughness. Location of the random roughness elements does not need to match that for subgrid roughness because the model attempts to simulate only the statistical effect of random roughness. Effect of individual roughness elements is insignificant as long as the generated roughness elements have the same statistical properties as the real subgrid roughness.

Downloaded From: http://fluidsengineering.asmedigitalcollection.asme.org/pdfaccess.ashx?url=/data/journals/jfega4/27181/ on 06/07/2017 Terms of Use: http://www.asme.org/al

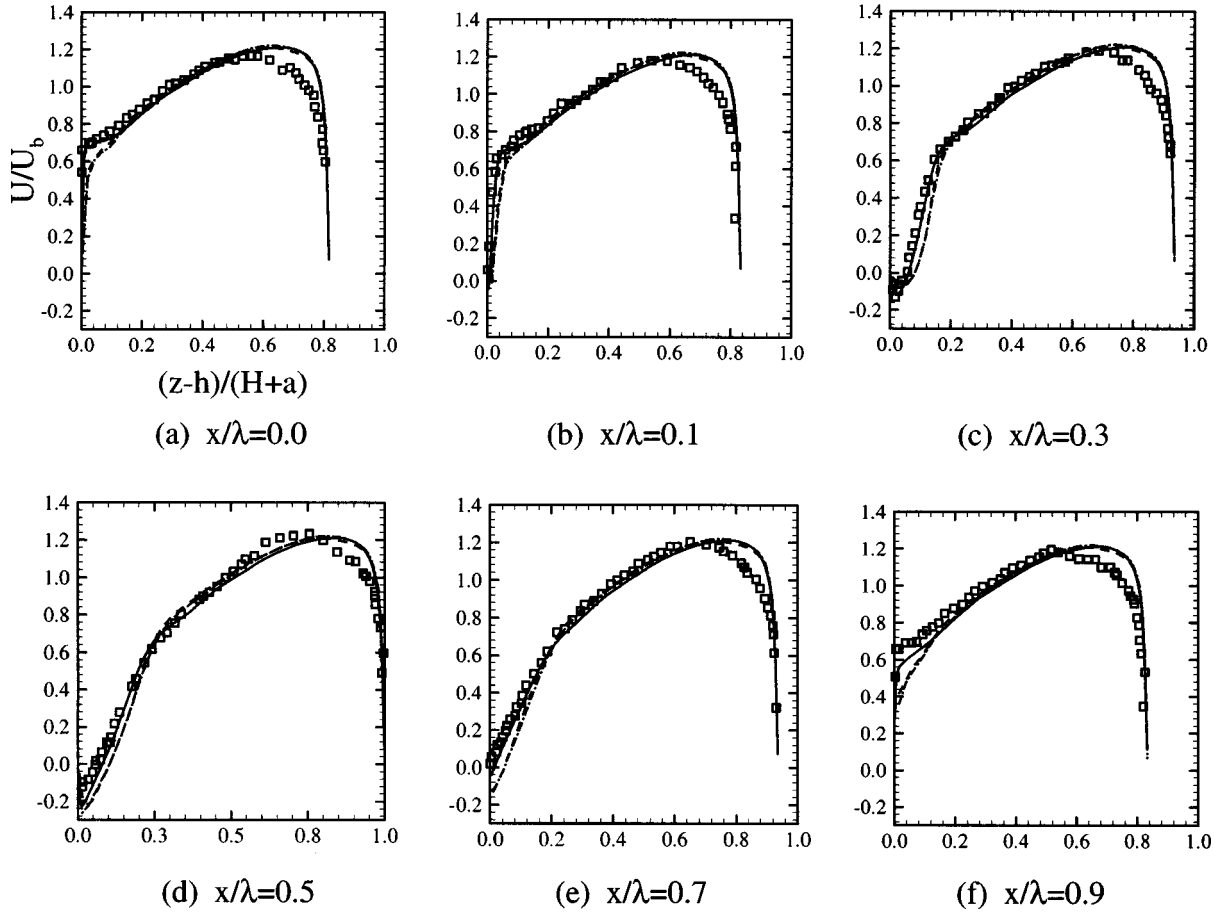


Fig. 7 Comparisons of mean streamwise velocity profiles using body-force model with experimental data at selected locations

Methods of generating random roughness elements are flexible. Height distribution of roughness elements can be simple random distribution, as in this study. Elements can be constructed on each grid point on the resolved roughness surface, or on every other grid point, or in other combinations, depending on the specific flow situation. Once random roughness elements are constructed, the body force is applied to grid points within the element.

The choice of the drag coefficient C_d needs further refinement. It is an empirical constant. Miyake et al. [11] chose 0.5 for channel flow, while Patton et al. [10] used a measured value of 0.4725 for a porous windbreak in boundary layer flow. The present example used 0.2. More numerical and laboratory experiments are needed to better correlate C_d and subgrid-scale roughness geometry. However, as subgrid roughness only accounts for the small-scale roughness, the value of C_d is unlikely to significantly affect the total flow field.

7 Summary and Conclusions

This paper has described a general method for treating the flow over a rough surface within the framework of large-eddy simulation. An arbitrary roughness is decomposed into two parts: resolved-scale and subgrid-scale roughness. The length scale of the resolved component is determined by the Cartesian grid that is chosen for a given flow situation, and is much larger than that of the subgrid roughness. For example, mountains can be treated as

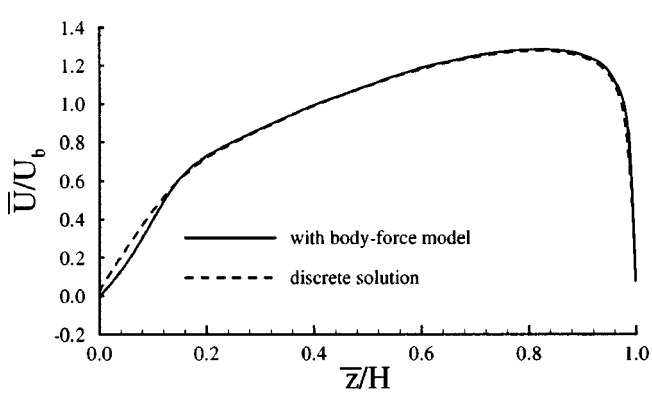


Fig. 8 Wave-averaged mean velocity profile

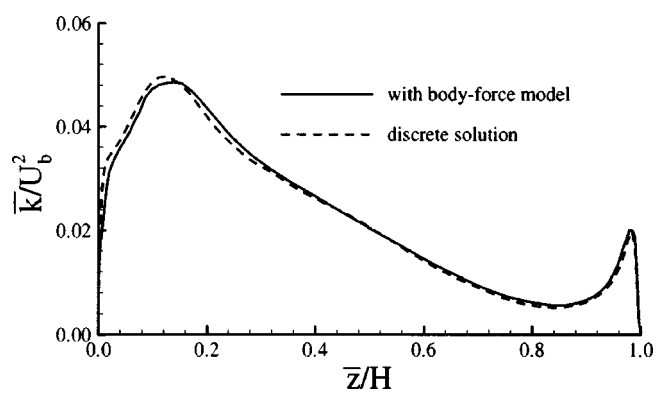


Fig. 9 Wave-averaged mean turbulence kinetic energy profile

Downloaded From: http://fluidsengineering.asmedigitalcollection.asme.org/pdfaccess.ashx?url=/data/journals/jfega4/27181/ on 06/07/2017 Terms of Use: http://www.asme.org/al

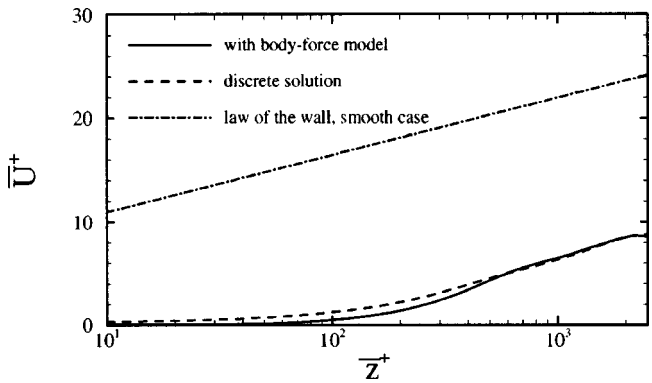


Fig. 10 Wave-averaged mean velocity profiles in wall coordinates

resolved-scale roughness, while trees and shrubs on the mountain surface are the subgrid roughness. The no-slip condition is realized by an immersed-boundary body force, which is internally determined and does not require a model. The resolved-scale roughness is accounted for directly, just as large-scale motions are solved directly in LES, and determines the major flow features. Such a simulation, by itself, may be sufficient for some applications.

For the subgrid roughness, a random body-force model is used. A drag coefficient has to be prescribed from known correlations or experiments. However, as the subgrid roughness is responsible for only a small portion of momentum, mass and energy transport, any uncertainty in this coefficient is not expected to be critical.

The new approach to modeling arbitrary roughness is demonstrated by the example of the flow over a wavy wall. Further details of turbulent flow over a wavy wall and application of the body-force model to flow over a rib roughness are given in Cui et al. [19].

Acknowledgments

The authors acknowledge the computer facilities and services provided by IIHR–Hydroscience and Engineering, National Center for Supercomputing Applications, and National Partnership for Advanced Computational Infrastructure. Prof. Robert Street and Dr. Yan Zang provided the source code developed at Stanford University.

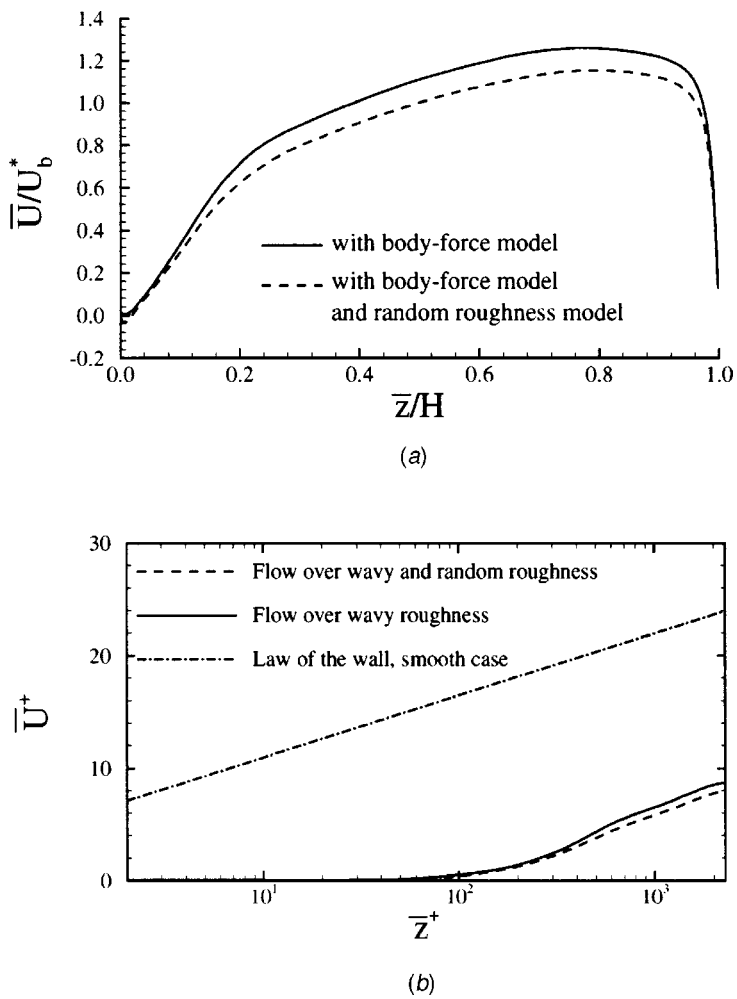


Fig. 11 Wave-averaged mean velocity profiles; effect of subgrid-scale roughness (a) In physical coordinates (U_b* is the bulk velocity in the case with body-force model only). (b) In logarithmic coordinates.

Downloaded From: http://fluidsengineering.asmedigitalcollection.asme.org/pdfaccess.ashx?url=/data/journals/jfega4/27181/ on 06/07/2017 Terms of Use: http://www.asme.org/al

References

[1] Nikuradse, J., 1950, ''Laws of Flow in Rough Pipes,'' National Advisory Committee for Aeronautics, NACA TM 1292 (translation from VDI-Forschungsheft 361, 1933).

[2] Adams, J. C., and Hodge, B. K., 1977, ''The Calculation of Compressible, Transitional, Turbulent and Relaminarizational Boundary Layers Over Smooth and Rough Surfaces Using an Extended Mixing Length Hypothesis,'' AIAA Paper No. 77-682.

[3] Lin, T. C., and Bywater, R. J., 1980, ''The Evaluation of Selected Turbulence Models for High-Speed Rough-Wall Boundary Layer Calculations,'' AIAA Paper No. 80-0132.

[4] Finson, M. L., 1982, ''A Model for Rough Wall Turbulent Heating and Skin Friction,'' AIAA Paper No. 82-0199.

[5] Christoph, G. H., and Pletcher, R. H., 1983, ''Prediction of Rough-Wall Skin Friction and Heat Transfer,'' AIAA J., **21**, pp. 509–515.

[6] Taylor, R. P., Coleman, H. W., and Hodge, B. K., 1985, ''Prediction of Turbulent Rough-Wall Skin Friction Using a Discrete Element Approach,'' ASME J. Fluids Eng., **107**, pp. 251–257.

[7] Maruyama, T., 1993, ''Optimization of Roughness Parameters for Staggered Arrayed Cubic Blocks Using Experimental Data,'' J. Wind. Eng. Ind. Aerodyn., **46**, pp. 165–171.

[8] Patel, V. C., 1998, ''Perspective: Flow at High Reynolds Number and Over Rough Surfaces—Achilles Heel of CFD,'' ASME J. Fluids Eng., **120**, pp. 434–444.

[9] Lin, C.-L., Moeng, C.-H., Sullivan, P. P., and McWilliams, J. C., 1997, ''The Effect of Surface Roughness on Flow Structures in a Neutrally Stratified Planetary Boundary Layer Flow,'' Phys. Fluids, **9**, pp. 3235–3249.

[10] Patton, E. G., Shaw, R. H., Judd, M. J., and Raupach, M. R., 1998, ''Large-Eddy Simulation of Windbreak Flow,'' Boundary-Layer Meteorol., **87**, pp. 275–306.

[11] Miyake, Y., Tsujimoto, K., and Agata, Y., 2000, ''A DNS of a Turbulent Flow in a Rough-Wall Channel Using Roughness Elements Model,'' JSME Int. J., **43**(2), pp. 233–242.

[12] Verzicco, R., Mohd-Yusof, J., Orlandi, P., and Haworth, D., 2000, ''Large Eddy Simulation in Complex Geometric Configurations Using Boundary Body Forces,'' AIAA J., **38**, pp. 427–433.

[13] Germano, M., Piomelli, U., Moin, P., and Cabot, W. H., 1991, ''A Dynamic Subgrid-Scale Eddy Viscosity Model,'' Phys. Fluids A, **3**, pp. 1760–1765.

[14] Piomelli, U., 1993, ''High Reynolds Number Calculations Using the Dynamic Subgrid-Scale Stress Model,'' Phys. Fluids, **5**, pp. 1484–1490.

[15] Lilly, D. K., 1992, ''A Proposed Modification of the Germano Subgrid-Scale Closure Method,'' Phys. Fluids, **4**, pp. 633–635.

[16] Zang, Y., Street, R. L., and Koseff, J. R., 1994, ''A Non-staggered Grid, Fractional Step Method for Time-Dependent Incompressible Navier-Stokes Equations in Curvilinear Coordinates,'' J. Comput. Phys., **114**, pp. 18–33.

[17] Buckles, T. J., Hanratty, T. J., and Adrian, R. J., 1984, ''Turbulent Flow Over Large-Amplitude Wavy Surfaces,'' J. Fluid Mech., **140**, pp. 27–44.

[18] Henn, D. S., and Sykes, R. I., 1999, ''Large-Eddy Simulation of Flow Over Wavy Surfaces,'' J. Fluid Mech., **383**, pp. 75–112.

[19] Cui, J., Patel, V. C., and Lin, C.-L., 2000, ''Large-Eddy Simulation of Turbulent Flow Over Rough Surfaces,'' IIHR Report No. 413, IIHR—Hydroscience & Engineering (formerly Iowa Institute of Hydraulic Research), The University of Iowa, Iowa City, IA.

[20] Fadlun, E. A., Verzicco, R., Orlandi, P., and Mohd-Yusof, J., 2000, ''Combined Immersed-Boundary Finite-Difference Methods for Three-Dimensional Complex Flow Simulations,'' J. Comput. Phys., **161**, pp. 35–60.

[21] Goldstein, D., Handler, R., and Sirovich, L., 1993, ''Modeling a No-Slip Flow Boundary With an External Force Field,'' J. Comput. Phys., **105**, pp. 354–366.

Downloaded From: http://fluidsengineering.asmedigitalcollection.asme.org/pdfaccess.ashx?url=/data/journals/jfega4/27181/ on 06/07/2017 Terms of Use: http://www.asme.org/al